

AI Interview Application

Business Requirements Document

Architect perspective with implementation-ready detail and workflow governance

Field	Value
Version	2.0
Prepared Date	2026-07-21
Document Type	Workflow-based enterprise BRD
Source Workflow	Registration to Offer/Rejection with fresher and experienced branches
Status	Draft for architecture, product, and operations sign-off

1. Executive Summary

The platform standardizes hiring execution from account registration to final offer or rejection. It combines AI-assisted screening with mandatory human gates, reducing manual delays while retaining decision accountability.

Architecture is modular and policy-driven: deterministic stage transitions for critical decisions, event-driven processing for long-running tasks, and immutable audit trails for compliance-sensitive actions.

2. Detailed Workflow

Step	Actor	Trigger	Output
Applicant Registration	Applicant	New applicant opens hiring portal	Verified candidate account
Login	Applicant, Recruiter, HR, Interviewer	Registered user submits credentials	Role-specific dashboard
Apply for Job	Applicant	Candidate selects open requisition	Draft or submitted application
Upload Resume	Applicant	Application in draft or submitted state	Resume stored and parsed profile
AI Resume Screening	System	Resume parsing complete	Screening score and recommendation
HR Review	HR Reviewer	AI result available	Shortlist or reject decision
Shortlisted Decision Branch	HR Reviewer	Review decision submitted	Workflow status updated
Rejection Communication	System	Not shortlisted	Communication log entry
Interview Process Initiation	Recruiter	Shortlisted candidate	Interview plan shell
Experience Check	Recruiter	Interview process initiated	Path assignment
Fresher Path Aptitude Test	Applicant	Path assigned as fresher	Aptitude score
Experienced Path Technical Interview	Technical Interviewer	Path assigned as experienced	Technical recommendation
Programming Assessment	Applicant	Aptitude or initial technical gate completed	Programming score
Technical Interview	Technical Interviewer	Programming score available	Technical panel recommendation
HR Round	HR Manager	Technical recommendation available	HR recommendation
Final Decision	HR Manager	All mandatory stages complete	Final decision record
Offer or Final Rejection	System and Recruiter	Final decision recorded	Candidate notified and workflow closed

3. Functional Requirement Modules

FR-REG: Registration, Login, and Candidate Identity

ID	Requirement	Priority	Acceptance
FR-REG-01	System shall support candidate self-registration with email or phone OTP verification.	High	User receives verified status and can log in.
FR-REG-02	System shall enforce configurable password policy and lockout thresholds.	High	After 5 failures, account lock event is logged.

ID	Requirement	Priority	Acceptance
FR-REG-03	System shall capture and version consent for privacy and AI processing notices.	High	Consent version appears in candidate audit timeline.
FR-REG-04	System shall support role-based login for applicant, recruiter, interviewer, HR manager, and admin.	High	Users only see permitted navigation and APIs.
FR-REG-05	System shall support optional SSO for internal users.	Medium	SSO users can authenticate without local password.

FR-APP: Job Search, Application, and Resume Capture

ID	Requirement	Priority	Acceptance
FR-APP-01	System shall display requisitions with eligibility criteria, status, and location.	High	Candidate can filter and apply to open jobs.
FR-APP-02	System shall prevent duplicate applications for the same requisition within a configurable cooling period.	High	Duplicate attempts show reason and cooling period end date.
FR-APP-03	System shall accept resume uploads in PDF and DOCX with malware scanning.	High	Upload returns accepted and scanning-complete status.
FR-APP-04	System shall parse resumes and map extracted fields to candidate profile attributes.	High	At least name, email, experience, skills, education are mapped.
FR-APP-05	System shall allow candidate edits to extracted profile before final submission.	Medium	Manual edits are audit-tracked separately from parser output.

FR-AI: AI Screening and Explainability

ID	Requirement	Priority	Acceptance
FR-AI-01	System shall compute match score using required skills, experience, and role fit.	High	Score and score-version are stored with timestamp.
FR-AI-02	System shall store top contributing factors and gaps for HR explainability view.	High	Reviewer sees positives, gaps, and confidence.
FR-AI-03	System shall flag low-confidence results for mandatory human review.	High	Low confidence routes candidate to manual queue.
FR-AI-04	System shall support configurable score thresholds per job family.	High	Threshold change effective date and author are logged.
FR-AI-05	System shall support model fallback mode when AI service is degraded.	Medium	System enters fallback state and notifies operations.

FR-REV: HR Review, Shortlist, and Routing

ID	Requirement	Priority	Acceptance
FR-REV-01	System shall provide HR review dashboard with candidate queue, filters, and aging indicators.	High	Queue can be filtered by requisition, stage, and SLA breach.
FR-REV-02	System shall require shortlist or reject decision with mandatory reason codes.	High	Decision submission blocked without required fields.
FR-REV-03	System shall trigger rejection communication automatically for not shortlisted candidates.	High	Communication event appears in timeline and audit logs.
FR-REV-04	System shall classify shortlisted candidates as fresher or experienced by policy.	High	Classified path visible and editable by authorized role.
FR-REV-05	System shall support recruiter override with justification and approval for exceptional routing.	Medium	Override request and approver are captured.

FR-INT: Assessments and Interviews

ID	Requirement	Priority	Acceptance
FR-INT-01	System shall schedule aptitude, coding, and interview stages with timezone-aware slots.	High	Candidate and panel receive correctly localized schedules.
FR-INT-02	System shall enforce fresher path with mandatory aptitude stage before coding.	High	Fresher cannot open coding link before aptitude completion.
FR-INT-03	System shall allow experienced path to begin with technical interview per policy.	High	Experienced candidate can proceed as configured.
FR-INT-04	System shall ingest aptitude and coding scores from external assessment provider.	High	Duplicate callback does not create duplicate score rows.
FR-INT-05	System shall capture interviewer scorecards with rubric sections and recommendation.	High	Scorecard includes mandatory rubric dimensions and comments.

ID	Requirement	Priority	Acceptance
FR-INT-06	System shall support no-show, reschedule, and cancellation states with reason codes.	Medium	State changes are tracked and visible in timeline.

FR-DEC: Final Decision, Offer Governance, and Closure

ID	Requirement	Priority	Acceptance
FR-DEC-01	System shall permit final decision only when mandatory stages are complete.	High	Decision button disabled until prerequisites are met.
FR-DEC-02	System shall support decision outcomes: offer, reject, hold, and withdraw.	High	Outcome reflected in candidate and requisition metrics.
FR-DEC-03	System shall enforce approval matrix for offer release based on compensation band.	High	Offer cannot be sent without required approvers.
FR-DEC-04	System shall produce auditable decision summary with stage scores and rationale.	High	Summary export available in recruiter console.
FR-DEC-05	System shall support candidate response tracking and auto-reminders for pending offers.	Medium	Reminder jobs run until response or expiry.

FR-COM: Communication and Notifications

ID	Requirement	Priority	Acceptance
FR-COM-01	System shall provide tokenized email templates for offer, rejection, interview invite, and reminders.	High	Template preview renders correctly with sample data.
FR-COM-02	System shall log all outbound communication with provider status.	High	Delivery, bounce, and retry statuses are visible.
FR-COM-03	System shall retry transient notification failures with exponential backoff.	High	Retries stop after max attempts and create task.
FR-COM-04	System shall support multi-channel alerts for internal users (email and in-app).	Medium	Recruiter receives both in-app alert and email for critical SLA events.
FR-COM-05	System shall support localization-ready template content.	Low	Template model includes locale and fallback locale.

4. Business Rules

Rule	If	Then
BR-01	Resume missing	Block screening
BR-02	Low AI confidence	Force HR review
BR-04	Fresher	Aptitude before coding
BR-06	Missing mandatory stage	Block final decision
BR-07	High compensation	Require extra approval
BR-10	Email bounce	Create recruiter task
BR-15	Deletion request	Anonymize with legal retention

5. UI and UX

- Dashboard-first workflow with explicit next action and SLA indicator.
- Accessible forms with keyboard support and readable error messaging.
- Candidate forms use autosave and progress indicators to reduce drop-off.
- No hidden status changes without explicit reviewer confirmation.

6. Architecture

Layer	Technology	Purpose
Frontend	React	Candidate and recruiter-facing web interfaces
Backend	Node.js	Core APIs and workflow orchestration
AI Integration	Python	AI screening and model integration services
Database	PostgreSQL	Primary relational data store

Component	Purpose	Constraint
Workflow Orchestrator	State transition control	Deterministic guards

Component	Purpose	Constraint
Resume and AI Service	Parse and score	Versioned model outputs
Assessment Adapter	External test sync	Idempotent callbacks
Decision Service	Final approval logic	Approval matrix enforcement
Audit Service	Immutable events	Append-only storage

7. Data and Integrations

Area	Details
Core Entities	Candidate, Application, Resume, Screening, Assessment, Feedback, Decision, Communication, Audit
ATS Integration	Requisition sync plus candidate status updates
Assessment Integration	Launch tests and ingest signed callbacks
Email Integration	Tokenized templates with provider status tracking
Retention	Decision and audit logs retained for compliance horizon

8. Non-Functional Targets

Category	Target
Latency	P95 < 2 seconds on core APIs under expected load
Availability	99.9 percent monthly uptime
Audit	100 percent decision path traceability
Security	No critical OWASP findings at release
Recovery	RPO <= 15 min, RTO <= 2 hours

9. Testing Strategy

Type	Scope	Owner
Unit	Business rules and stage guards	Engineering
Integration	ATS, assessment, email, calendar	Engineering QA
E2E	Both candidate branches and edge states	QA and Product
Security	Authz and data protection controls	Security QA
UAT	Operational readiness and usability	HR Ops

Automation testing framework: Playwright with TypeScript for cross-browser E2E validation, regression, and CI execution.

10. Risks

Risk	Mitigation
AI misclassification	Human review gate and threshold tuning
Provider outage	Retry queue and manual fallback
Data access incident	RBAC hardening and incident playbook
SLA breaches	Escalation dashboards and queue balancing

11. Roles and Access

Role	Core Responsibilities
Applicant	Register, apply, upload documents, complete assessments
Recruiter	Manage funnel, shortlist operations, communications
HR Reviewer	Validate screening outputs and progression decisions
Technical Interviewer	Run interviews and submit rubric scorecards
HR Manager	Finalize decision and approvals
System Admin	Configuration, access controls, integration health

12. Communication Templates

Includes offer, rejection, interview invitation, and reminder templates with tokenized placeholders.

13. Compliance and Governance

- Least-privilege access with API-level authorization checks.
- Encrypted PII storage and secure transport.
- Immutable audit records for shortlist and final decision actions.
- Retention and deletion aligned to legal and policy requirements.

14. Delivery Roadmap

Milestone	Outcome
M1	Identity and application flow live in SIT
M2	Screening and HR review production-ready
M3	Interview orchestration and decision controls complete
M4	Communications, reporting, and go-live readiness signed off

15. Open Issues and Glossary

Open items include path classification policy, compensation approval thresholds, and provider finalization.
Glossary covers ATS, SLA, RBAC, RPO, RTO, and idempotency terms for implementation alignment.